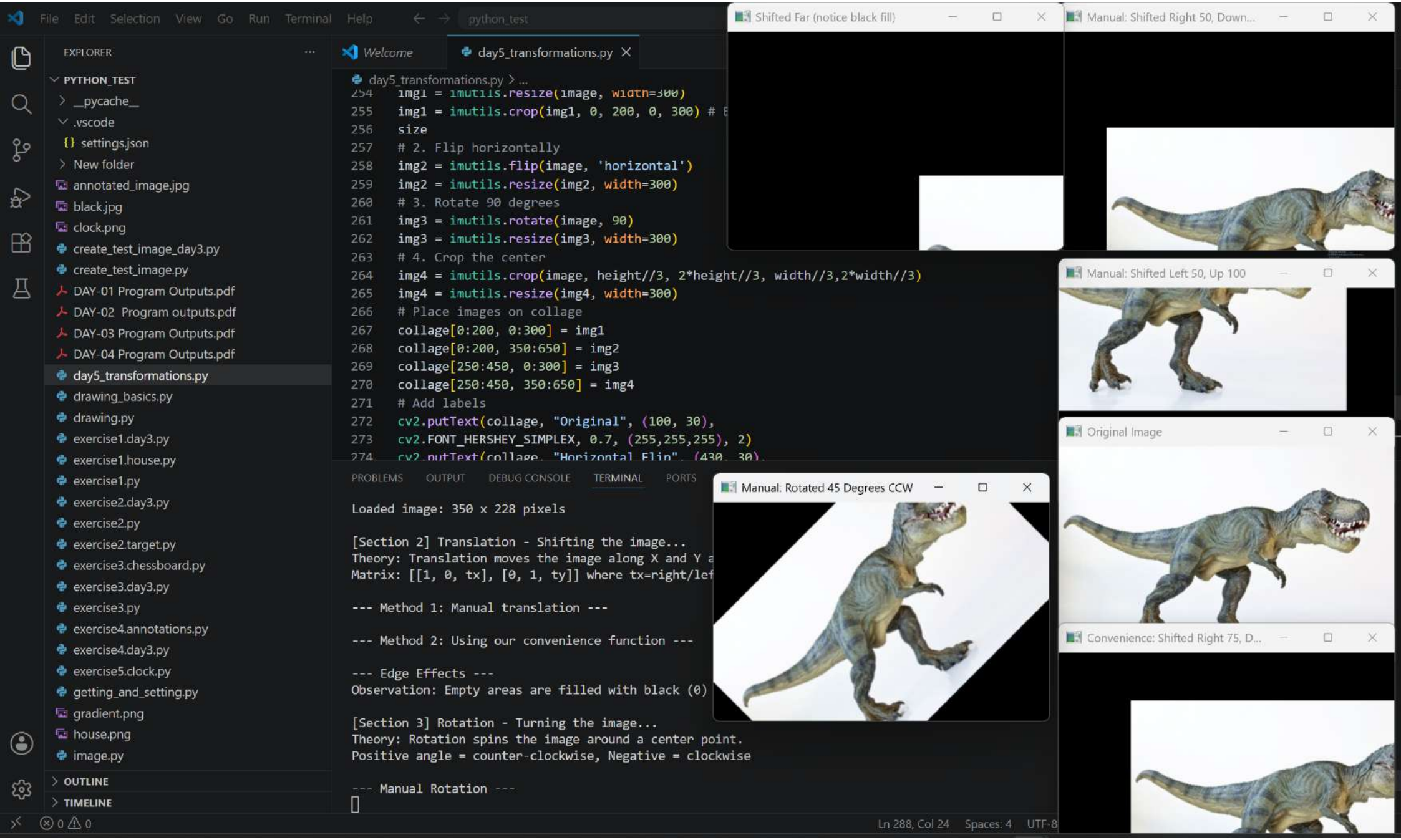
File Edit Selection View Go Run Terminal Help
python_test
EXPLORER
PYTHON_TEST
__pycache__
.vscode
settings.json
New folder
annotated_image.jpg
black.jpg
clock.png
create_test_image_day3.py
create_test_image.py
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DAY-02 Program outputs.pdf
DAY-03 Program Outputs.pdf
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drawing.py
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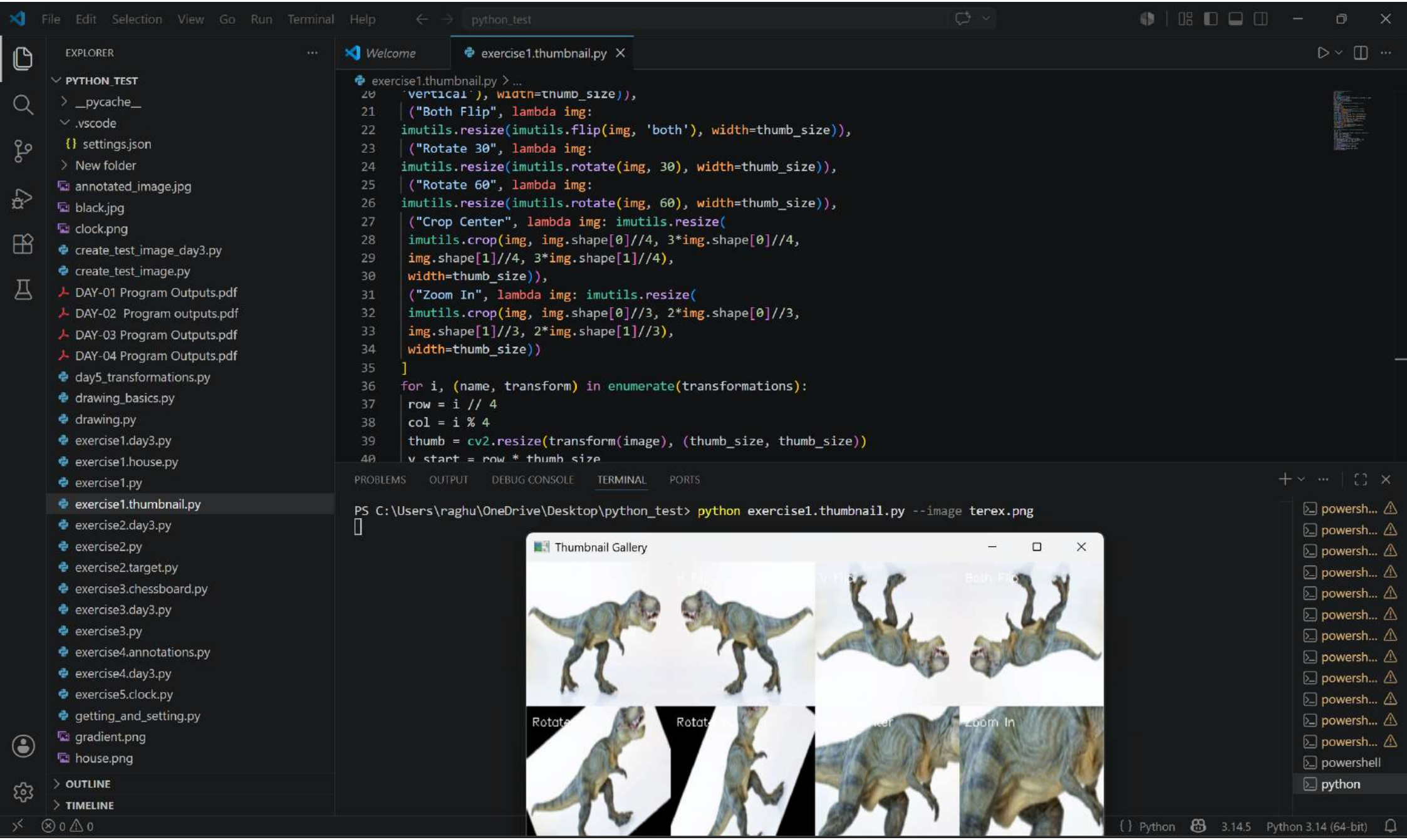
exercise3.day3.py
exercise3.py
exercise4.annotations.py
exercise4.day3.py
exercise5.clock.py
getting_and_setting.py
gradient.png
house.png
image.py
OUTLINE
TIMELINE
Welcome
day5_transformations.py
day5_transformations.py > ...
254 img1 = imutils.resize(image, width=300)
255 img1 = imutils.crop(img1, 0, 200, 0, 300) # E
256 size
257 # 2. Flip horizontally
258 img2 = imutils.flip(image, 'horizontal')
259 img2 = imutils.resize(img2, width=300)
260 # 3. Rotate 90 degrees
261 img3 = imutils.rotate(image, 90)
262 img3 = imutils.resize(img3, width=300)
263 # 4. Crop the center
264 img4 = imutils.crop(image, height//3, 2*height//3, width//3,2*width//3)
265 img4 = imutils.resize(img4, width=300)
266 # Place images on collage
267 collage[0:200, 0:300] = img1
268 collage[0:200, 350:650] = img2
269 collage[250:450, 0:300] = img3
270 collage[250:450, 350:650] = img4
271 # Add labels
272 cv2.putText(collage, "Original", (100, 30),
273 cv2.FONT_HERSHEY_SIMPLEX, 0.7, (255,255,255), 2)
274 cv2.putText(collage, "Horizontal Flip", (430, 30),
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Loaded image: 350 x 228 pixels
[Section 2] Translation - Shifting the image...
Theory: Translation moves the image along X and Y a
Matrix: [[1, 0, tx], [0, 1, ty]] where tx=right/lef
--- Method 1: Manual translation ---
--- Method 2: Using our convenience function ---
--- Edge Effects ---
Observation: Empty areas are filled with black (0)
[Section 3] Rotation - Turning the image...
Theory: Rotation spins the image around a center point.
Positive angle = counter-clockwise, Negative = clockwise
--- Manual Rotation ---
Ln 288, Col 24 Spaces: 4 UTF-8
Shifted Far (notice black fill)
Manual: Shifted Right 50, Down...
Manual: Shifted Left 50, Up 100
Original Image
Manual: Rotated 45 Degrees CCW
Convenience: Shifted Right 75, D...


File Edit Selection View Go Run Terminal Help
python_test
EXPLORER
PYTHON_TEST
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New folder
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black.jpg
clock.png
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create_test_image.py
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exercise1.py
exercise1.thumbnail.py
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exercise2.target.py
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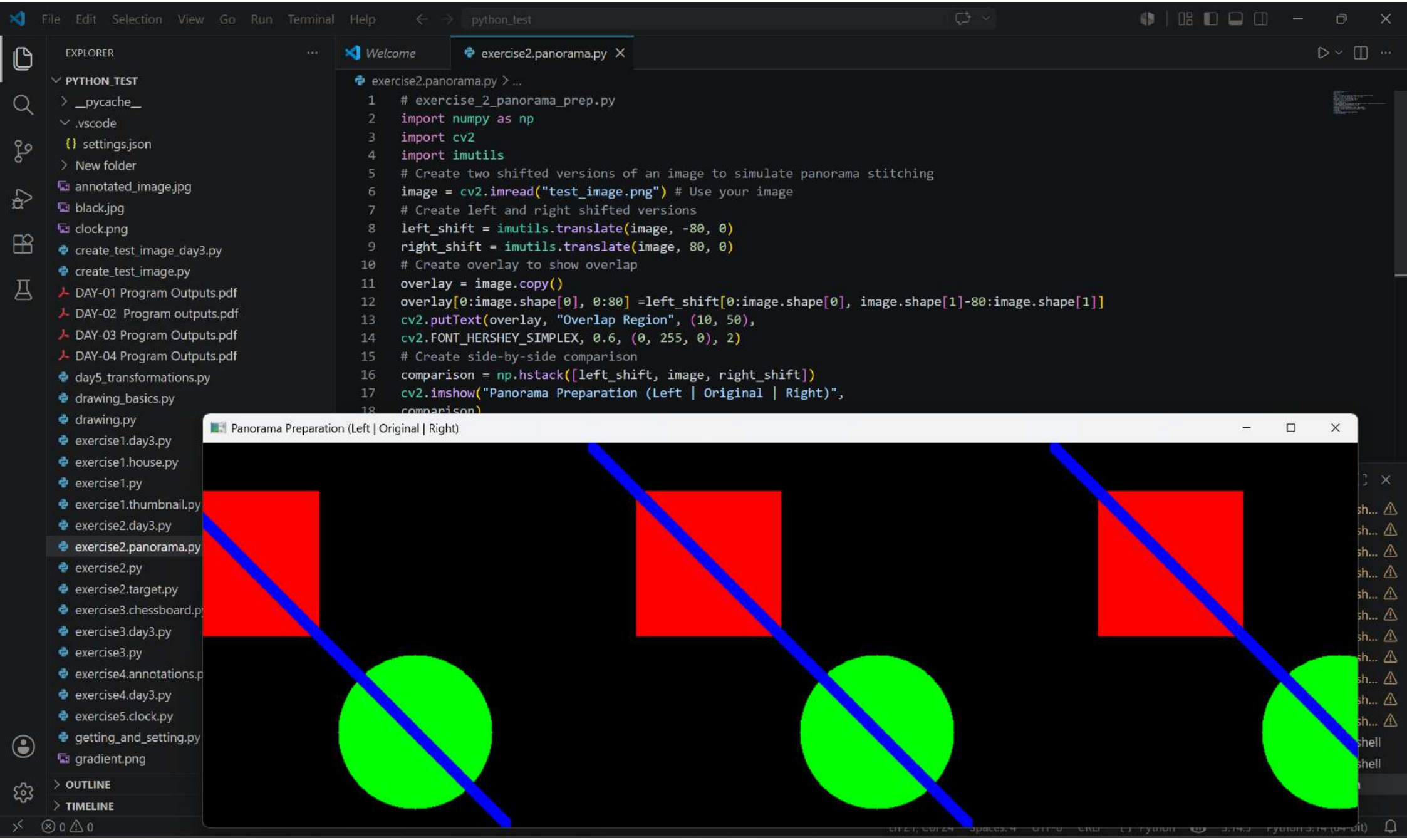
exercise3.day3.py
exercise3.py
exercise4.annotations.py
exercise4.day3.py
exercise5.clock.py
getting_and_setting.py
gradient.png
house.png
OUTLINE
TIMELINE
Welcome
exercise1.thumbnail.py
exercise1.thumbnail.py > ...
("Both Flip", lambda img:
imutils.resize(imutils.flip(img, 'both'), width=thumb_size)),
("Rotate 30", lambda img:
imutils.resize(imutils.rotate(img, 30), width=thumb_size)),
("Rotate 60", lambda img:
imutils.resize(imutils.rotate(img, 60), width=thumb_size)),
("Crop Center", lambda img: imutils.resize(
imutils.crop(img, img.shape[0]//4, 3*img.shape[0]//4,
img.shape[1]//4, 3*img.shape[1]//4),
width=thumb_size)),
("Zoom In", lambda img: imutils.resize(
imutils.crop(img, img.shape[0]//3, 2*img.shape[0]//3,
img.shape[1]//3, 2*img.shape[1]//3),
width=thumb_size))
]
for i, (name, transform) in enumerate(transformations):
row = i // 4
col = i % 4
thumb = cv2.resize(transform(image), (thumb_size, thumb_size))
y_start = row * thumb_size
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise1.thumbnail.py --image terex.png
Thumbnail Gallery
Zoom In
powersh...
powershell
python
Python 3.14.5 Python 3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
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exercise1.py
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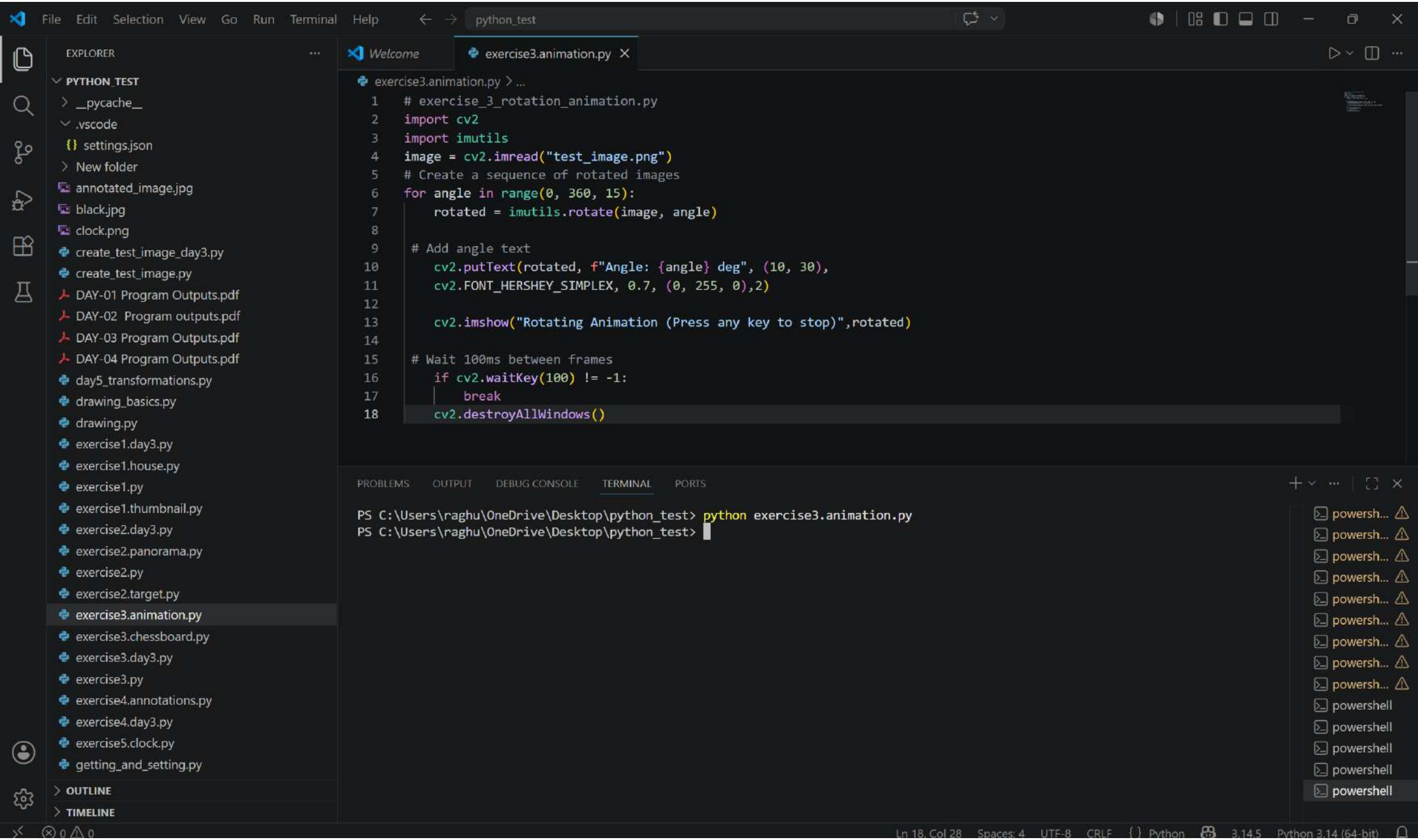
exercise2.panorama.py
exercise2.py
exercise2.target.py
exercise3.chessboard.py
exercise3.day3.py
exercise3.py
exercise4.annotations.p
exercise4.day3.py
exercise5.clock.py
getting_and_setting.py
gradient.png
OUTLINE
TIMELINE
Welcome
exercise2.panorama.py
exercise2.panorama.py > ...
1 # exercise_2_panorama_prep.py
2 import numpy as np
3 import cv2
4 import imutils
5 # Create two shifted versions of an image to simulate panorama stitching
6 image = cv2.imread("test_image.png") # Use your image
7 # Create left and right shifted versions
8 left_shift = imutils.translate(image, -80, 0)
9 right_shift = imutils.translate(image, 80, 0)
10 # Create overlay to show overlap
11 overlay = image.copy()
12 overlay[0:image.shape[0], 0:80] =left_shift[0:image.shape[0], image.shape[1]-80:image.shape[1]]
13 cv2.putText(overlay, "Overlap Region", (10, 50),
14 cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0, 255, 0), 2)
15 # Create side-by-side comparison
16 comparison = np.hstack([left_shift, image, right_shift])
17 cv2.imshow("Panorama Preparation (Left | Original | Right)",
Panorama Preparation (Left | Original | Right)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exercise3.chessboard.py
exercise3.day3.py
exercise3.py
exercise4.annotations.py
exercise4.day3.py
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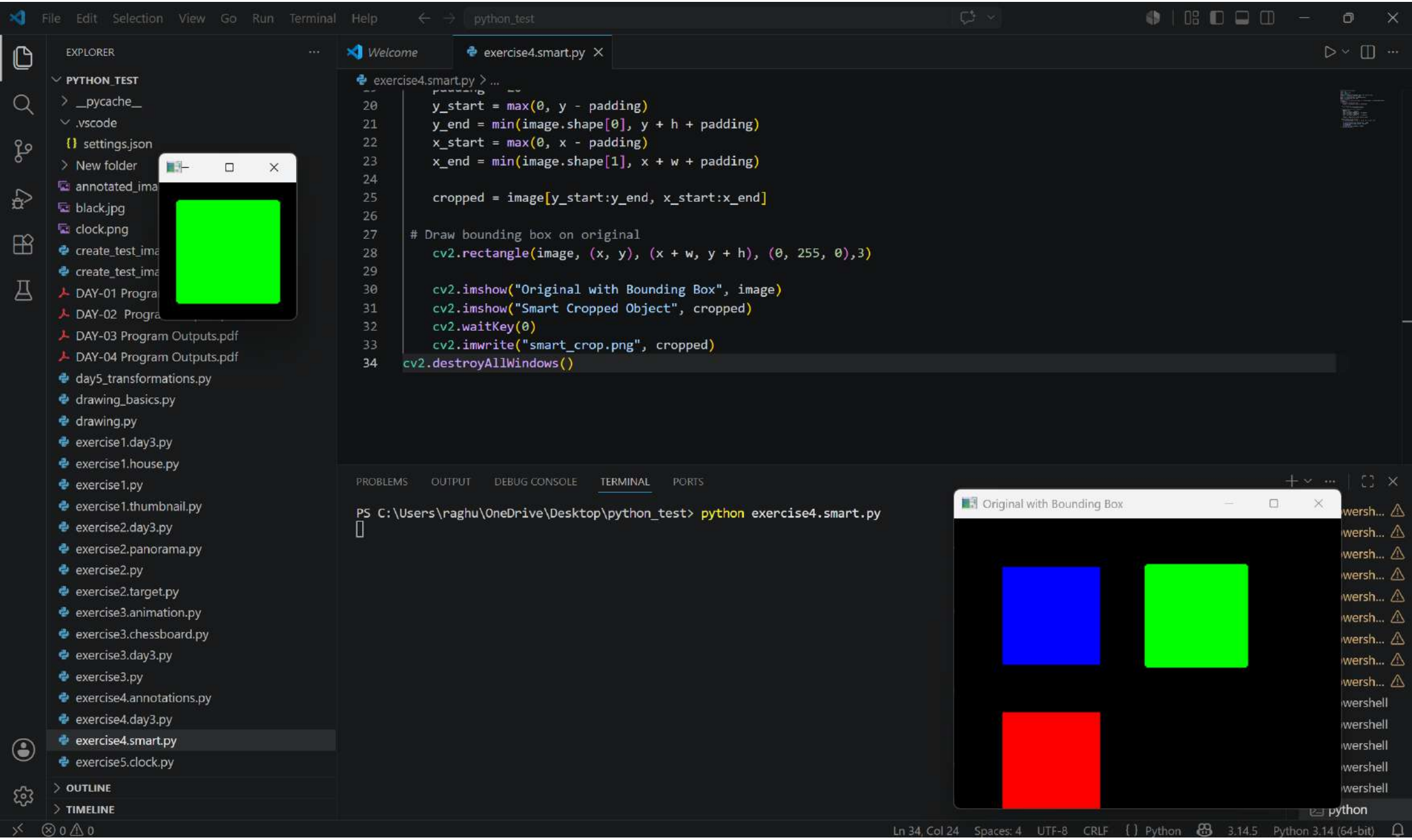
getting_and_setting.py
OUTLINE
TIMELINE
Welcome
exercise3.animation.py
exercise3.animation.py > ...
1 # exercise_3_rotation_animation.py
2 import cv2
3 import imutils
4 image = cv2.imread("test_image.png")
5 # Create a sequence of rotated images
6 for angle in range(0, 360, 15):
7 rotated = imutils.rotate(image, angle)
8
9 # Add angle text
10 cv2.putText(rotated, f"Angle: {angle} deg", (10, 30),
11 cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 0),2)
12
13 cv2.imshow("Rotating Animation (Press any key to stop)",rotated)
14
15 # Wait 100ms between frames
16 if cv2.waitKey(100) != -1:
17 break
18 cv2.destroyAllWindows()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise3.animation.py
PS C:\Users\raghu\OneDrive\Desktop\python_test>
powersh...
powershell
Ln 18, Col 28 Spaces: 4 UTF-8 CRLF Python 3.14.5 Python 3.14 (64-bit)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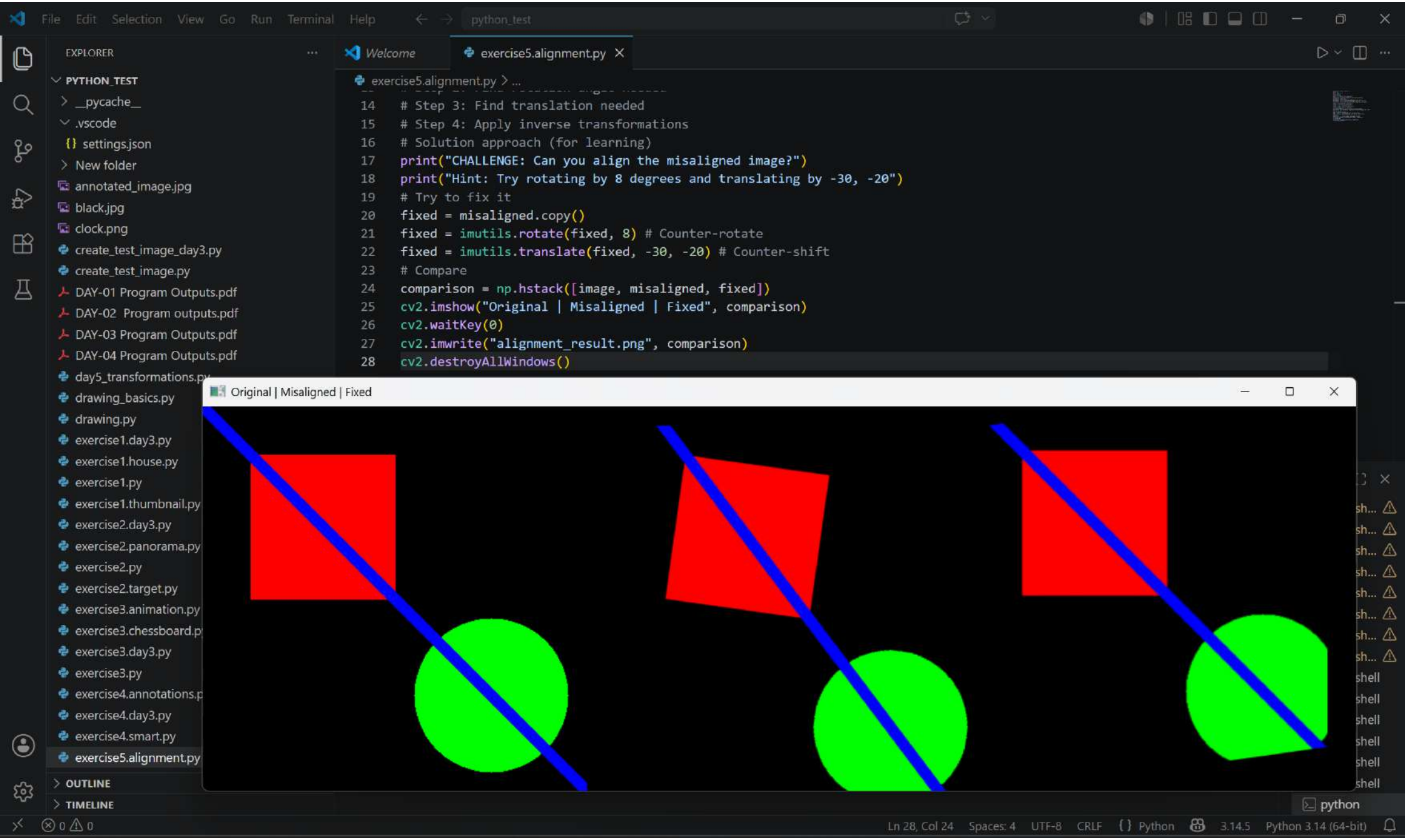
exercise3.chessboard.py
exercise3.day3.py
exercise3.py
exercise4.annotations.py
exercise4.day3.py
exercise4.smart.py
exercise5.clock.py
OUTLINE
TIMELINE
Welcome
exercise4.smart.py
exercise4.smart.py > ...
20 y_start = max(0, y - padding)
21 y_end = min(image.shape[0], y + h + padding)
22 x_start = max(0, x - padding)
23 x_end = min(image.shape[1], x + w + padding)
24
25 cropped = image[y_start:y_end, x_start:x_end]
26
27 # Draw bounding box on original
28 cv2.rectangle(image, (x, y), (x + w, y + h), (0, 255, 0),3)
29
30 cv2.imshow("Original with Bounding Box", image)
31 cv2.imshow("Smart Cropped Object", cropped)
32 cv2.waitKey(0)
33 cv2.imwrite("smart_crop.png", cropped)
34 cv2.destroyAllWindows()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise4.smart.py
Original with Bounding Box
python
Ln 34, Col 24 Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
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exercise1.house.py
exercise1.py
exercise1.thumbnail.py
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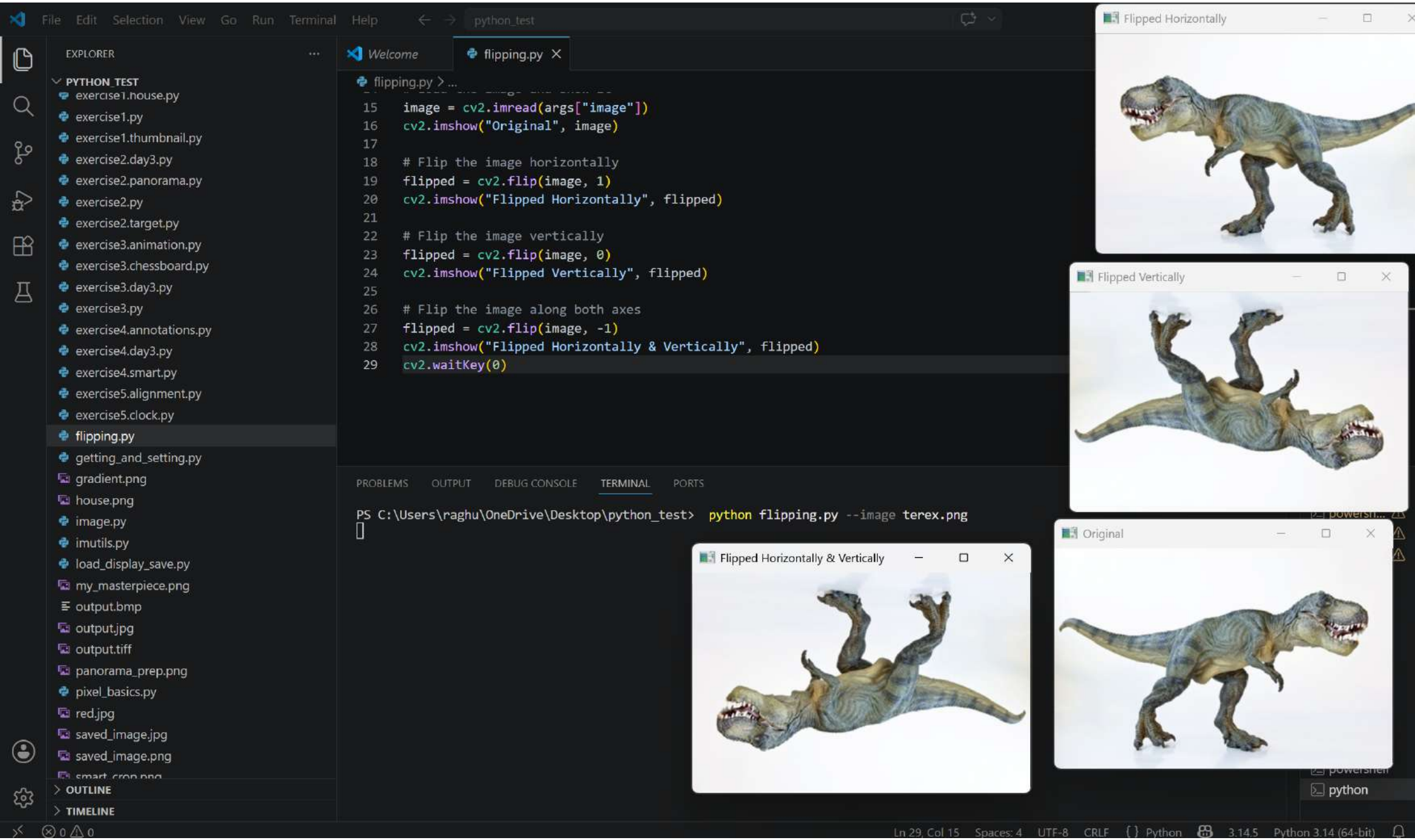
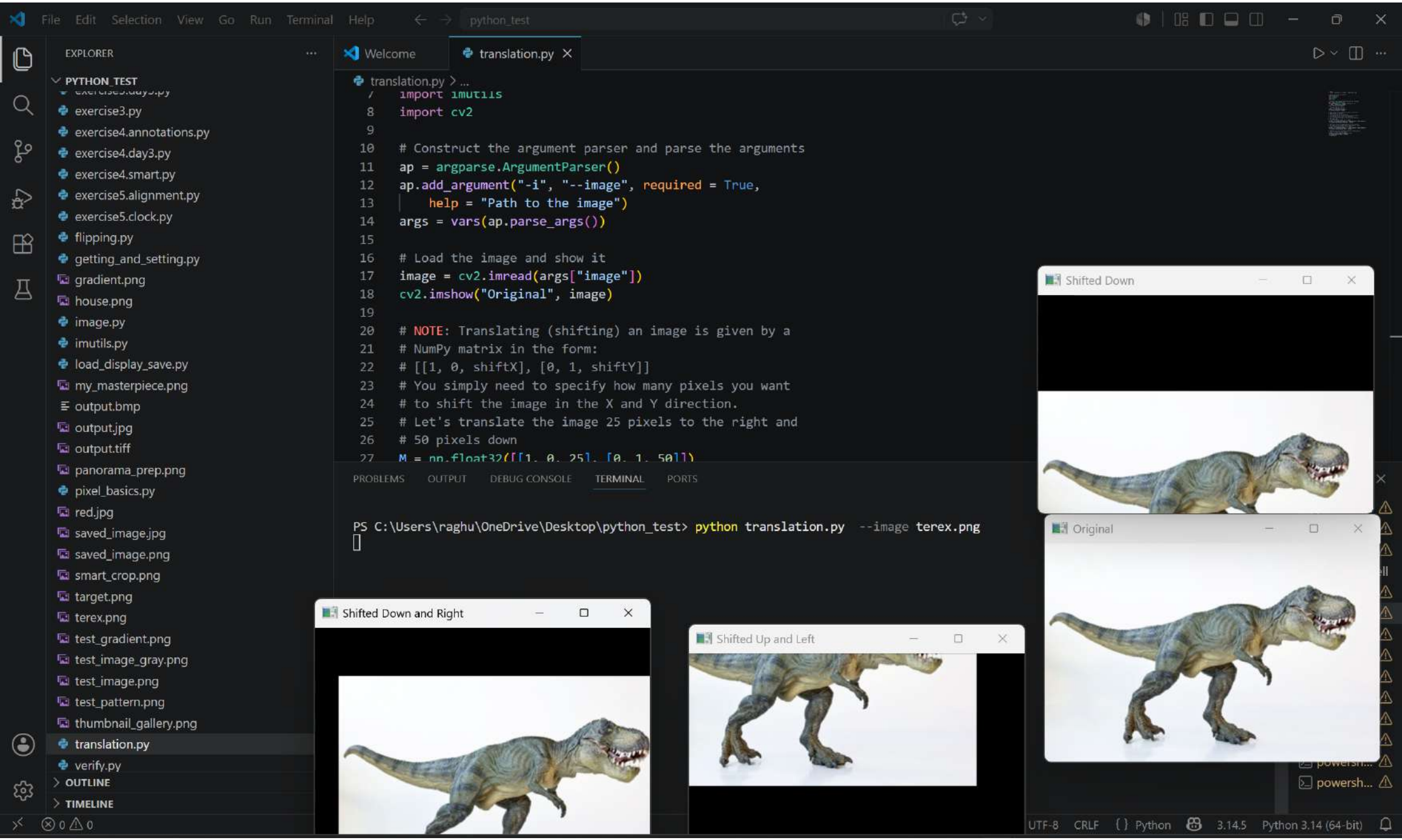
exercise2.panorama.py
exercise2.py
exercise2.target.py
exercise3.animation.py
exercise3.chessboard.p
exercise3.day3.py
exercise3.py
exercise4.annotations.p
exercise4.day3.py
exercise4.smart.py
exercise5.alignment.py
OUTLINE
TIMELINE
Welcome
exercise5.alignment.py
exercise5.alignment.py > ...
14 # Step 3: Find translation needed
15 # Step 4: Apply inverse transformations
16 # Solution approach (for learning)
17 print("CHALLENGE: Can you align the misaligned image?")
18 print("Hint: Try rotating by 8 degrees and translating by -30, -20")
19 # Try to fix it
20 fixed = misaligned.copy()
21 fixed = imutils.rotate(fixed, 8) # Counter-rotate
22 fixed = imutils.translate(fixed, -30, -20) # Counter-shift
23 # Compare
24 comparison = np.hstack([image, misaligned, fixed])
25 cv2.imshow("Original | Misaligned | Fixed", comparison)
26 cv2.waitKey(0)
27 cv2.imwrite("alignment_result.png", comparison)
28 cv2.destroyAllWindows()
Original | Misaligned | Fixed
python
Ln 28, Col 24 Spaces: 4 UTF-8 CRLF Python 3.14.5 Python 3.14 (64-bit)

```
# USAGE
# python crop.py --image ../images/trex.png

# Import the necessary packages
import numpy as np
import argparse
import cv2

# Construct the argument parser and parse the arguments
ap = argparse.ArgumentParser()
ap.add_argument("-i", "--image", required = True,
    help = "Path to the image")
args = vars(ap.parse_args())

# Load the image and show it
image = cv2.imread(args["image"])
cv2.imshow("Original", image)

# Cropping an image is as simple as using array slices
# in NumPy! Let's crop out the face of the T-Rex. The
```

```
PS C:\Users\raghu\OneDrive\Desktop\python_test> python crop.py --image terex.png
```


File Edit Selection View Go Run Terminal Help
python_test
EXPLORER
PYTHON_TEST
exercise1.house.py
exercise1.py
exercise1.thumbnail.py
exercise2.day3.py
exercise2.panorama.py
exercise2.py
exercise2.target.py
exercise3.animation.py
exercise3.chessboard.py
exercise3.day3.py
exercise3.py
exercise4.annotations.py
exercise4.day3.py
exercise4.smart.py
exercise5.alignment.py
exercise5.clock.py
flipping.py
getting_and_setting.py
gradient.png
house.png
image.py
imutils.py
load_display_save.py
my_masterpiece.png
output.bmp
output.jpg
output.tiff
panorama_prep.png
pixel_basics.py
red.jpg
saved_image.jpg
saved_image.png
OUTLINE
TIMELINE
Welcome
flipping.py
flipping.py > ...
15 image = cv2.imread(args["image"])
16 cv2.imshow("Original", image)
17
18 # Flip the image horizontally
19 flipped = cv2.flip(image, 1)
20 cv2.imshow("Flipped Horizontally", flipped)
21
22 # Flip the image vertically
23 flipped = cv2.flip(image, 0)
24 cv2.imshow("Flipped Vertically", flipped)
25
26 # Flip the image along both axes
27 flipped = cv2.flip(image, -1)
28 cv2.imshow("Flipped Horizontally & Vertically", flipped)
29 cv2.waitKey(0)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python flipping.py --image terex.png
Flipped Horizontally
Flipped Vertically
Flipped Horizontally & Vertically
Original
python
Ln 29, Col 15 Spaces: 4 UTF-8 CRLF Python 3.14.5 Python 3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
python_test
EXPLORER
PYTHON_TEST
exercise3.py
exercise4.annotations.py
exercise4.day3.py
exercise4.smart.py
exercise5.alignment.py
exercise5.clock.py
flipping.py
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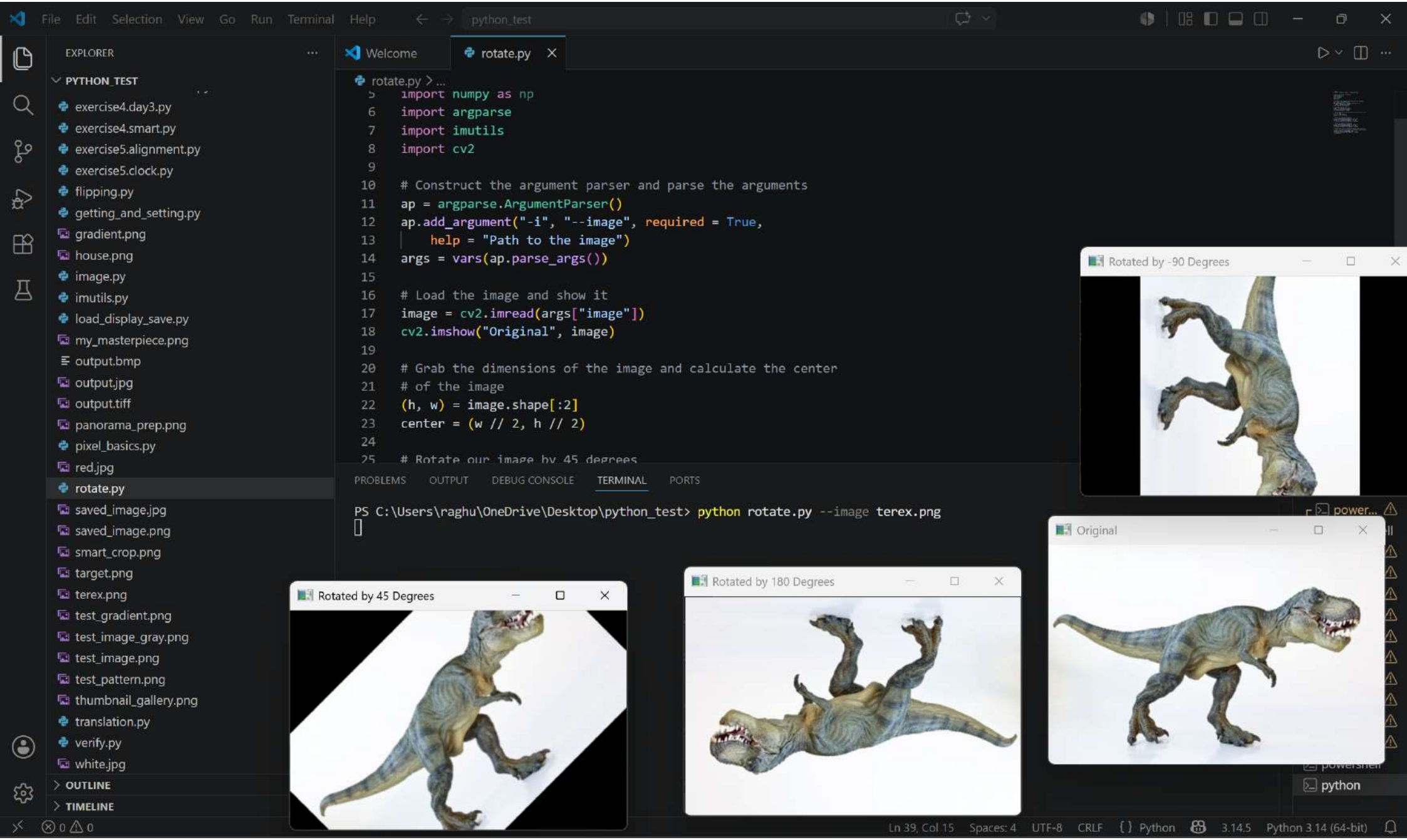
gradient.png
house.png
image.py
imutils.py
load_display_save.py
my_masterpiece.png
output.bmp
output.jpg
output.tiff
panorama_prep.png
pixel_basics.py
red.jpg
saved_image.jpg
saved_image.png
smart_crop.png
target.png
terex.png
test_gradient.png
test_image_gray.png
test_image.png
test_pattern.png
thumbnail_gallery.png
translation.py
verify.py
OUTLINE
TIMELINE
Welcome
translation.py
translation.py > ...
import imutils
import cv2
# Construct the argument parser and parse the arguments
ap = argparse.ArgumentParser()
ap.add_argument("-i", "--image", required = True,
help = "Path to the image")
args = vars(ap.parse_args())
# Load the image and show it
image = cv2.imread(args["image"])
cv2.imshow("Original", image)
# NOTE: Translating (shifting) an image is given by a
# NumPy matrix in the form:
# [[1, 0, shiftX], [0, 1, shiftY]]
# You simply need to specify how many pixels you want
# to shift the image in the X and Y direction.
# Let's translate the image 25 pixels to the right and
# 50 pixels down
M = np.float32([[1, 0, 25], [0, 1, 50]])
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python translation.py --image terex.png
Shifted Down
Original
Shifted Down and Right
Shifted Up and Left
powersh...
powersh...
UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
python_test
EXPLORER
PYTHON_TEST
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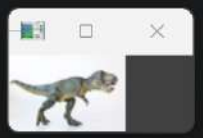
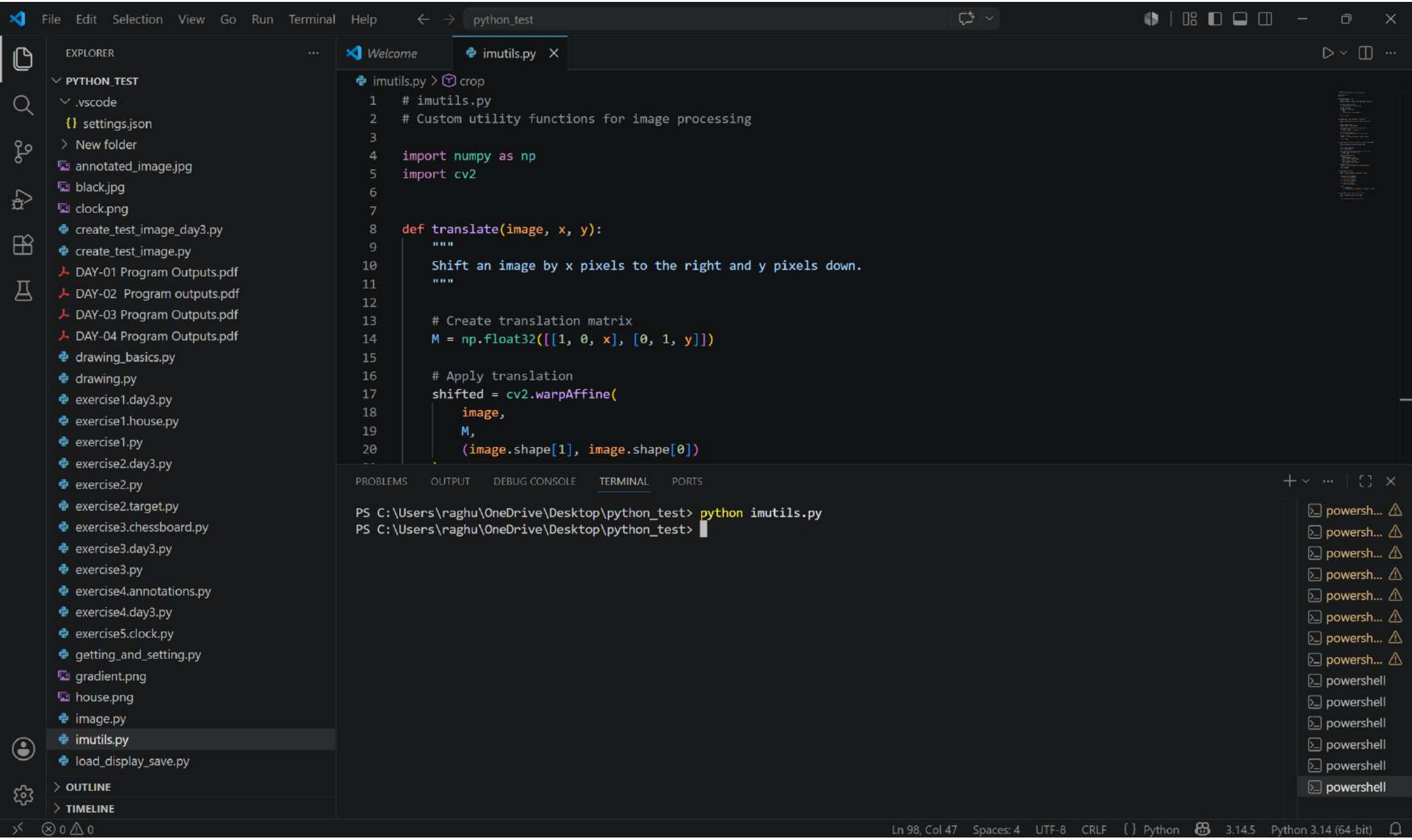
output.jpg
output.tiff
panorama_prep.png
pixel_basics.py
red.jpg
rotate.py
saved_image.jpg
saved_image.png
smart_crop.png
target.png
terex.png
test_gradient.png
test_image_gray.png
test_image.png
test_pattern.png
thumbnail_gallery.png
translation.py
verify.py
white.jpg
OUTLINE
TIMELINE
Welcome
rotate.py
rotate.py > ...
import numpy as np
import argparse
import imutils
import cv2
# Construct the argument parser and parse the arguments
ap = argparse.ArgumentParser()
ap.add_argument("-i", "--image", required = True,
help = "Path to the image")
args = vars(ap.parse_args())
# Load the image and show it
image = cv2.imread(args["image"])
cv2.imshow("Original", image)
# Grab the dimensions of the image and calculate the center
# of the image
(h, w) = image.shape[:2]
center = (w // 2, h // 2)
# Rotate our image by 45 degrees
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python rotate.py --image terex.png
Rotated by -90 Degrees
Rotated by 45 Degrees
Rotated by 180 Degrees
Original
power...
python
Ln 39, Col 15 Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)

```
# USAGE
# python resize.py --image ../images/trex.png

# Import the necessary packages
import numpy as np
import argparse
import imutils
import cv2

# Construct the argument parser and parse the arguments
ap = argparse.ArgumentParser()
ap.add_argument("-i", "--image", required = True,
    help = "Path to the image")
args = vars(ap.parse_args())

# Load the image and show it
image = cv2.imread(args["image"])
cv2.imshow("Original", image)

# We need to keep in mind aspect ratio so the image does
```

```
PS C:\Users\raghu\OneDrive\Desktop\python_test> python resize.py --image terex.png
```

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exercise3.py
exercise4.annotations.py
exercise4.day3.py
exercise5.clock.py
getting_and_setting.py
gradient.png
house.png
image.py
imutils.py
load_display_save.py
OUTLINE
TIMELINE
Welcome
imutils.py
imutils.py > crop
1 # imutils.py
2 # Custom utility functions for image processing
3
4 import numpy as np
5 import cv2
6
7
8 def translate(image, x, y):
9 """
10 Shift an image by x pixels to the right and y pixels down.
11 """
12
13 # Create translation matrix
14 M = np.float32([[1, 0, x], [0, 1, y]])
15
16 # Apply translation
17 shifted = cv2.warpAffine(
18 image,
19 M,
20 (image.shape[1], image.shape[0])
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python imutils.py
PS C:\Users\raghu\OneDrive\Desktop\python_test>
powersh...
powershell
Ln 98, Col 47 Spaces: 4 UTF-8 CRLF Python 3.14.5 Python 3.14 (64-bit)